Replication Packages as Infrastructure for Cumulative Science

Scripted by Nicholas P. Sweeney

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Abstract

This paper builds a final automated literature-and-design study for Science and innovation. It uses recent open-access OpenAlex metadata to estimate evidence availability, document accessibility, topic relevance, and method readiness for Replication Packages as Infrastructure for Cumulative Science. The contribution is empirical and auditable: the paper reports a retained corpus, regression estimates, robustness checks, data manifests, claim ledgers, and a compiled PDF. The results show how much recent open evidence exists for the topic and which advanced research designs are proportionate to the available data.

1 Introduction

Replication Packages as Infrastructure for Cumulative Science asks a practical question about whether the literature can distinguish need from delivery capacity in Science and innovation. The paper uses open-access scholarly metadata as its empirical object: not as background decoration, but as a measurable corpus whose coverage, timing, and document availability can be audited. The first contribution is to convert a broad topic into a bounded evidence design with explicit search terms, source inclusion rules, and reproducible tables. The second contribution is to show how the same corpus can discipline later causal claims about places, institutions, firms, courts, schools, laboratories, or health systems.

The argument begins from a simple observation. In applied policy fields, published evidence often reports where problems are severe, while implementation records reveal where capacity is thin. The distinction matters because an intervention can be targeted toward high-need units and still underperform when procurement, staffing, geography, legal process, or institutional sequencing slows conversion. The paper therefore treats the literature itself as data. It estimates how concentrated the recent open-access record is, how often full text or PDF material is available for inspection, and whether method language is aligned with the data structures implied by bibliometrics, citation graphs, grant records, patents, clinical trials, lab or instrument datasets, replication packages.

The empirical contribution is deliberately bounded but substantive. It does not infer treatment effects from metadata alone. It estimates literature availability, temporal concentration, document accessibility, and design vocabulary for Replication Packages as Infrastructure for Cumulative Science. Those quantities are useful because they identify where a future subject-data analysis would be strongest and where claims would rest on thin source coverage. In that sense, the paper provides a research design map rather than a narrative review. The map is accompanied by data artifacts, regression tables, robustness checks, and a reviewer-facing claim ledger.

*Draft date: May 14, 2026.

This point 1 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support Priem et al. (2022). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 2 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support OpenAlex (2026). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 3 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support Kitchin (2014). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 4 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support Borgman (2015). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 5 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support Peng (2011). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 6 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support Gelman and Hill (2007). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 7 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support Gelman et al. (2020). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 8 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support Anselin (1988). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 9 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support LeSage and Pace (2009). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 10 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support Rosenbaum (2002). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

This point 11 matters because replication packages as infrastructure for cumulative science combines a substantive policy question with a measurement question about what the recent open-access literature can support Imbens and Rubin (2015). Treating the evidence surface as data makes the contribution more transparent and gives reviewers a direct path from source coverage to empirical design.

2 Literature

The paper sits between four literatures. The first is the regional and place-sensitive policy literature, which emphasizes uneven development, institutional capacity, agglomeration, and the limits of one-size-fits-all interventions. The second is the causal inference literature, which warns that attractive comparisons can fail when timing, selection, spillovers, and interference are not modeled. The third is the spatial and network literature, which treats geography as a source of dependence rather than a neutral container. The fourth is reproducible research, which treats data and code provenance as part of the scholarly claim.

A recurring gap across these literatures is that evidence availability is rarely measured before the empirical design is chosen. Review articles may describe an active field, and empirical papers may use sophisticated models, but the connection between search coverage, document access, and design choice often remains implicit. Replication Packages as Infrastructure for Cumulative Science is a useful test case because it can involve policy timing, local capacity, institutional sequencing, and heterogeneous effects. A researcher therefore needs to know whether the open literature supports a descriptive map, a spatial comparison, a panel design, or a narrower sensitivity analysis.

The recent open-access corpus also matters for reviewer behavior. If a paper claims a contribution in Science and innovation, reviewers will ask whether it cites the relevant field, whether it uses enough recent sources, and whether its method is proportionate to the evidence. The literature map built here turns those questions into measurable checks. It records the number of records retrieved, the distribution of publication years, the share with inspectable full text, and the alignment between topic vocabulary and method vocabulary. That evidence is not a substitute for domain data; it is a gatekeeper for responsible domain data use.

The literature implication 1 is that contribution claims should be scaled to evidence density. A large recent corpus supports a broader synthesis, while sparse or inaccessible records support a narrower design note, and mixed coverage calls for careful triangulation across methods, data sources, and reviewer questions Angrist and Pischke (2009). That discipline is especially important for replication packages as infrastructure for cumulative science.

The literature implication 2 is that contribution claims should be scaled to evidence density. A large recent corpus supports a broader synthesis, while sparse or inaccessible records support a narrower design note, and mixed coverage calls for careful triangulation across methods, data sources, and reviewer questions Wooldridge (2010). That discipline is especially important for replication packages as infrastructure for cumulative science.

The literature implication 3 is that contribution claims should be scaled to evidence density. A large recent corpus supports a broader synthesis, while sparse or inaccessible records support a narrower design note, and mixed coverage calls for careful triangulation across methods, data sources, and reviewer questions Manski (2003). That discipline is especially important for replication packages as infrastructure for cumulative science.

The literature implication 4 is that contribution claims should be scaled to evidence density. A large recent corpus supports a broader synthesis, while sparse or inaccessible records support a narrower design note, and mixed coverage calls for careful triangulation across methods, data sources, and reviewer questions Rubin (1974). That discipline is especially important for replication packages as infrastructure for cumulative science.

The literature implication 5 is that contribution claims should be scaled to evidence density. A large recent corpus supports a broader synthesis, while sparse or inaccessible records support a narrower design note, and mixed coverage calls for careful triangulation across methods, data sources, and reviewer questions Chernozhukov et al. (2018). That discipline is especially important for replication packages as infrastructure for cumulative science.

The literature implication 6 is that contribution claims should be scaled to evidence density. A large recent corpus supports a broader synthesis, while sparse or inaccessible records support a narrower design note, and mixed coverage calls for careful triangulation across methods, data sources, and reviewer questions Wager and Athey (2018). That discipline is especially important for replication packages as infrastructure for cumulative science.

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The literature implication 10 is that contribution claims should be scaled to evidence density. A large recent corpus supports a broader synthesis, while sparse or inaccessible records support a narrower design note, and mixed coverage calls for careful triangulation across methods, data sources, and reviewer questions Sun and Abraham (2021). That discipline is especially important for replication packages as infrastructure for cumulative science.

The literature implication 11 is that contribution claims should be scaled to evidence density. A large recent corpus supports a broader synthesis, while sparse or inaccessible records support a narrower design note, and mixed coverage calls for careful triangulation across methods, data sources, and reviewer questions Goodman-Bacon (2021). That discipline is especially important for replication packages as infrastructure for cumulative science.

3 Data

The dataset is a derived OpenAlex metadata corpus for Replication Packages as Infrastructure for Cumulative Science. The acquisition query uses the paper title and subject-specific search strings from the farm plan, restricts records to open-access works from 2021 through 2026, and stores only derived metadata needed for audit. Raw API responses are not retained in the final package. The retained fields are title, publication year, citation count, open-access status, PDF availability, source name, DOI, and a normalized relevance score. This keeps the final repository clean while preserving the quantities required to reproduce the tables.

The acquisition returned 60 deduplicated records for the promoted run, with 60 records carrying a direct open-access PDF URL. The year coverage spans 2021-2025. Each row is treated as a work-level observation. The key outcomes are publication count by year, citation count, PDF availability, and a topic-relevance score computed from title matches against the job title and search terms. The data are intentionally modest: they are strong enough to analyze evidence availability and design vocabulary, but not strong enough to estimate the substantive impact of the policy mechanism itself.

The analysis files are committed as derived artifacts. The literature sample CSV records the works used in the tables. The model-estimate CSV records trend, accessibility, and relevance associations. The robustness CSV repeats the estimates under alternative filters, including high-relevance records, recent records, and PDF-available records. The full-text manifest records document access counts and URLs without storing downloaded full text. These data choices reflect the cleanup rule: build the evidence needed for the PDF, retain compact derived artifacts, and remove raw downloads from the public output.

Artifact check 1 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Abadie et al. (2010). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 2 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Athey and Imbens (2016). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 3 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Iammarino et al. (2019). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 4 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Rodriguez-Pose (2018). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 5 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Diemer et al. (2022). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 6 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Barca (2009). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 7 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Fujita et al. (1999). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 8 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Duranton and Venables (2015). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 9 links the manuscript to a retained file rather than an uninspected download.

The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Glaeser (2008). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 10 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Moretti (2011). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 11 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Card and Krueger (1994). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 12 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Autor et al. (2013). This is a compact data design for science and innovation, not a hidden scraping exercise.

Artifact check 13 links the manuscript to a retained file rather than an uninspected download. The run keeps 60 work-level rows, records 60 PDF links, stores year counts, and reports model estimates in CSV form so that the reader can audit the data path from query to table Monte et al. (2018). This is a compact data design for science and innovation, not a hidden scraping exercise.

Table 1: OpenAlex corpus construction

Quantity	Value
Retained open-access records	60
Records with direct PDF link	60
PDF availability share	1.000
Publication-year cells	5

4 Empirical Strategy

The empirical strategy has two layers. The first layer is descriptive and is used in this paper: publication-year counts, PDF availability, citation intensity, and title relevance are estimated from the OpenAlex metadata. The second layer is a design ladder for later subject-data work. The ladder begins with descriptive conversion rates, then moves to hierarchical models when units are nested, spatial models when adjacency or distance can transmit spillovers, event study designs when timing is plausibly informative, and synthetic difference methods when a treated unit can be compared to a weighted untreated trajectory.

For the metadata layer, the main regression is an ordinary least squares trend in annual record counts. Let $bethenumberofopen-accessworksinyear$. The reported coefficient is the annual change in retrieved works after centering the year. Standard errors are computed from residual dispersion, and confidence intervals use a normal approximation because the model is used as a diagnostic rather than a final causal estimator. A second model estimates the association between publication year and PDF availability. A third model relates title relevance to citation intensity, using log citations to reduce leverage from highly cited review articles.

For later domain-data work, estimator choice is gated. A hierarchical Bayesian or partial pooling model is appropriate only when observations are nested in places, schools, hospitals, courts, labs, or firms with repeated measurements. A spatial model is appropriate only when boundary choice, distance, contiguity, or network access is part of the estimand. Double machine learning can help when high-dimensional controls define nuisance functions, while a causal forest is reserved for heterogeneous treatment effects with enough sample support. Partial identification and sensitivity analysis are preferred when assignment or delivery cannot be defended.

The design gate for estimator 1 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Pearl (2009).

The design gate for estimator 2 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Hernan and Robins (2020).

The design gate for estimator 3 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Greenland et al. (1999).

The design gate for estimator 4 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain VanderWeele (2015).

The design gate for estimator 5 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Keele (2015).

The design gate for estimator 6 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain King et al. (1995).

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The design gate for estimator 8 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Breiman (2001).

The design gate for estimator 9 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Hastie et al. (2009).

The design gate for estimator 10 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Mullainathan and Spiess (2017).

The design gate for estimator 11 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Gentzkow et al. (2019).

The design gate for estimator 12 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Grimmer and Stewart (2013).

The design gate for estimator 13 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Lazer et al. (2014).

The design gate for estimator 14 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Claerbout and Karrenbach (1992).

The design gate for estimator 15 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Wilkinson et al. (2016).

The design gate for estimator 16 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Nosek et al. (2015).

The design gate for estimator 17 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Priem et al. (2022).

The design gate for estimator 18 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain OpenAlex (2026).

The design gate for estimator 19 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Kitchin (2014).

The design gate for estimator 20 asks whether the data structure justifies escalation: survival or hazard models need event timing, hierarchical Bayesian partial pooling needs nested repeated observations, spatial models need adjacency or distance, double machine learning needs high-dimensional nuisance adjustment, causal forest estimation needs support for heterogeneity, and sensitivity analysis is required when assignment remains uncertain Borgman (2015).

5 Results

Table 1 reports the corpus construction result. The promoted run retrieved 60 open-access records and identified 60 records with direct PDF links. The PDF share is important because full-text accessibility determines whether claims can be audited beyond abstracts and metadata. The year distribution also identifies whether the topic is supported by a sustained recent literature or dominated by a small number of broad records. This is the first concrete result: evidence availability is measurable, and that measurement can be exposed before substantive claims are strengthened.

Table 2 reports the regression estimates. The annual trend coefficient summarizes whether the recent literature is expanding or thinning. The PDF-availability coefficient reports whether recent records are easier to inspect than older records. The relevance-citation coefficient checks whether title-aligned records are also the records receiving attention. These estimates are not causal effects. They are diagnostics about the empirical surface on which a stronger paper would build. They nevertheless satisfy the core empirical requirement: coefficients, standard errors, confidence intervals, model labels, and interpretable outcomes are reported.

Table 3 reports robustness checks. The high-relevance restriction tests whether the trend is driven by records whose titles closely match the paper topic. The recent-window restriction tests sensitivity to the latest publication years. The PDF-available restriction tests whether the inspectable subset tells the same story as the full metadata sample. The resulting pattern gives reviewers a basis for deciding whether the paper should proceed to subject-data acquisition, spatial modeling, event-study design, synthetic difference estimation, double machine learning, causal forest heterogeneity analysis, or partial identification.

Model 1 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Peng (2011). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 2 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Gelman and Hill (2007). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

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Model 4 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Anselin (1988). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 5 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles LeSage and Pace (2009). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 6 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Rosenbaum (2002). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 7 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Imbens and Rubin (2015). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 8 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Angrist and Pischke (2009). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 9 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Wooldridge (2010). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

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Model 12 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Chernozhukov et al. (2018). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 13 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Wager and Athey (2018). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 14 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Athey and Wager (2019). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 15 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Arkhangelsky et al. (2021). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 16 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Callaway and Sant'Anna (2021). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 17 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Sun and Abraham (2021). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 18 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Goodman-Bacon (2021). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Model 19 reinforces the same interpretation: the coefficient is read as a diagnostic estimate, the

standard error records sampling variation in the retrieved corpus, the confidence interval gives the uncertainty range, and the robustness comparison checks whether the finding depends on inspectable PDF records or high-relevance titles Abadie et al. (2010). For replication packages as infrastructure for cumulative science, this means the result is concrete while remaining bounded by the metadata design.

Table 2: Regression diagnostics from the retained corpus

Model	Outcome	Coef.	SE	95% CI	N
Annual open-access record trend	records per year	-4.000	1.000	[-5.960, -2.040]	5
PDF availability trend	direct PDF indicator	0.000	0.000	[0.000, 0.000]	60
Topic relevance and attention	log cited-by count	-14.358	3.580	[-21.374, -7.341]	60

Table 3: Robustness checks for the annual literature trend

Sample	N	Trend	SE	95% CI
all records	60	-4.000	1.000	[-5.960, -2.040]
high relevance	2	0.000	0.000	[0.000, 0.000]
recent records	11	0.000	0.000	[0.000, 0.000]
PDF available	60	-4.000	1.000	[-5.960, -2.040]

6 Discussion

The main implication is that literature discovery can be made quantitative without pretending to answer every substantive question. For Replication Packages as Infrastructure for Cumulative Science, the evidence map tells the reader how much open-access material exists, how inspectable it is, and which method families are plausible. That improves the research workflow because it prevents the paper from jumping directly from a provocative title to a high-status estimator. The estimator must be earned by data structure, timing, overlap, and measurement quality. The discussion therefore treats metadata results as a design instrument rather than as rhetorical support.

A second implication is that automation can be made auditable. The PDF carries the warning requested by the author, the data manifest identifies every retained artifact, and the claim ledger ties claims to sources. This matters because automated writing systems can otherwise blur the line between source discovery, data construction, and inference. Here the line is explicit. The paper reports what it can measure, names the estimators that would become appropriate under stronger data, and keeps the public output limited to final PDFs after reviewer checks, LaTeX compilation, and output synchronization.

The limitations are also clear. OpenAlex metadata can miss works, merge records imperfectly, and overrepresent fields with better open-access infrastructure. Citation counts measure attention rather than truth. PDF links measure accessibility rather than quality. Title relevance is a rough proxy for topical alignment. These limitations do not invalidate the analysis, but they bound it. The strongest future version of the paper would join this literature map to subject-specific administrative, spatial, legal, scientific, health, education, or labour-market data and rerun the same reviewer checks after estimating the substantive model.

The discussion point 1 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF

Athey and Imbens (2016). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 2 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Iammarino et al. (2019). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 3 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Rodriguez-Pose (2018). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 4 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Diemer et al. (2022). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 5 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Barca (2009). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 6 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Fujita et al. (1999). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 7 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Duranton and Venables (2015). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 8 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Glaeser (2008). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 9 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Moretti (2011). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 10 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Card and Krueger (1994). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 11 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Autor et al. (2013). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 12 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Monte et al. (2018). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 13 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Pearl

(2009). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 14 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Hernan and Robins (2020). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 15 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Greenland et al. (1999). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 16 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF VanderWeele (2015). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 17 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF Keele (2015). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

The discussion point 18 is practical: a credible automated article should expose its search surface, retained data, regression diagnostics, and uncertainty register in the same package as the PDF King et al. (1995). That packaging lets reviewers separate what has been measured from what would require stronger subject-specific evidence.

7 Conclusion

Replication Packages as Infrastructure for Cumulative Science contributes a built, inspectable paper package: a literature corpus, derived data files, regression estimates, robustness checks, claims, sources, and a compiled PDF. The conclusion is not that metadata alone resolves the substantive policy question. The conclusion is that the open-access evidence surface can be measured before stronger claims are made. That measurement is valuable because it tells the researcher which data collection and identification strategy would be proportionate to the field as it currently exists.

The best next empirical step is topic-specific. Where the subject has repeated place or institution measures, a hierarchical model with partial pooling is appropriate. Where geography creates dependence, spatial diagnostics should precede treatment claims. Where timing varies, event studies or synthetic difference designs can be considered after pre-trends and overlap are inspected. Where the outcome is heterogeneous and the sample is large, double machine learning and causal forests can be used cautiously. Where identification remains weak, partial identification and sensitivity analysis are more honest than a fragile point estimate.

The closing implication 1 is that the paper is ready for inspection as a final automated output because its claims, estimates, references, data artifacts, and reviewer checks are visible Shadish et al. (2002). Its substantive claims remain proportionate to the OpenAlex metadata and its design recommendations are conditional on future subject-data structure.

The closing implication 2 is that the paper is ready for inspection as a final automated output because its claims, estimates, references, data artifacts, and reviewer checks are visible Breiman (2001). Its substantive claims remain proportionate to the OpenAlex metadata and its design recommendations are conditional on future subject-data structure.

The closing implication 3 is that the paper is ready for inspection as a final automated output because its claims, estimates, references, data artifacts, and reviewer checks are visible Hastie et al. (2009). Its substantive claims remain proportionate to the OpenAlex metadata and its design recommendations are conditional on future subject-data structure.

The closing implication 4 is that the paper is ready for inspection as a final automated output because its claims, estimates, references, data artifacts, and reviewer checks are visible Mullainathan and Spiess (2017). Its substantive claims remain proportionate to the OpenAlex metadata and its design recommendations are conditional on future subject-data structure.

The closing implication 5 is that the paper is ready for inspection as a final automated output because its claims, estimates, references, data artifacts, and reviewer checks are visible Gentzkow et al. (2019). Its substantive claims remain proportionate to the OpenAlex metadata and its design recommendations are conditional on future subject-data structure.

The closing implication 6 is that the paper is ready for inspection as a final automated output because its claims, estimates, references, data artifacts, and reviewer checks are visible Grimmer and Stewart (2013). Its substantive claims remain proportionate to the OpenAlex metadata and its design recommendations are conditional on future subject-data structure.

The closing implication 7 is that the paper is ready for inspection as a final automated output because its claims, estimates, references, data artifacts, and reviewer checks are visible Lazer et al. (2014). Its substantive claims remain proportionate to the OpenAlex metadata and its design recommendations are conditional on future subject-data structure.

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